Measuring the Gap: Semantic Alignment Between AI-for-Sustainability Research and SDG Policy Frameworks

Contents

Abstract

The Sustainable Development Goals (SDGs) are the dominant international development framework. Shared SDG labels do not necessarily imply shared semantic framing between research and policy. Existing bibliometric studies map AI research to SDGs using keyword or classification methods (Armitage et al., 2020; Kashnitsky et al., 2024), but rarely compare the semantic content of research and policy discourse. This study uses Sentence-BERT embeddings and a supervised logistic regression classifier (test macro-F1 = 0.823) to map semantic distance between 3,105,144 AI-for-sustainability abstracts and 43,147 segments from 2,455 institutional SDG policy sources. The analysis separates coverage gaps (which SDGs each corpus emphasises) from within-SDG semantic gaps (how far texts assigned to the same SDG sit apart). Because the corpora differ in register (Biber and Conrad, 2009), these gaps are observed textual-semantic divergence, not substantive disagreement.

The results find: First, policy discourse concentrates on SDGs 16 (Peace and Strong Institutions, 38.9%), 13 (Climate Action, 20.2%), and 17 (Partnerships, 19.0%), together 78.1% of policy coverage. Second, research concentrates on SDGs 3 (Good Health, 25.3%), 9 (Industry and Infrastructure, 18.3%), and 4 (Quality Education, artefact-affected, 14.6%). Third, the largest within-SDG semantic gaps occur in SDGs 13 (0.478), 3 (0.459), 11 (0.456), 1 (0.443), and 7 (0.433). Fourth, research coverage alone does not predict semantic divergence (H1c: $r = 0.062$, $p = 0.814$; $n = 17$) — this dissociation between attention and framing is the study's most robust finding. Fifth, H1a, the paper's primary hypothesis, tests whether the coverage gap predicts semantic divergence: the association is near-zero in the canonical specification ($r = -0.066$, $p = 0.801$), and no hypothesis survives robustness checks across alternative assignment methods, policy source families, or segment caps — coverage and framing are independent dimensions of divergence, not linked effects.

These findings do not prove substantive research-policy misalignment. They show that shared SDG labels are insufficient evidence of shared semantic framing, and that embedding-based distance is a transparent diagnostic map of proximity and divergence. The results need caution because of register effects, policy source-family heterogeneity, OpenAlex retrieval boundaries, and an SDG 4 assignment that is artefact-affected rather than a clean education finding.

1 Introduction

Artificial intelligence is now routinely presented as part of the solution to sustainable development. In academic research, papers increasingly claim that a model, application, or dataset contributes to one or more Sustainable Development Goals (SDGs), with SDG attribution often functioning as a form of relevance-signalling beyond technical novelty. In international institutional SDG policy discourse, international organisations, intergovernmental bodies, and national AI strategies likewise frame AI as an enabler of SDG progress (OECD, 2019; UK Government, 2021; UN AI Advisory Body, 2024; UNESCO, 2021; United Nations, 2015). Across these domains, a shared narrative has taken hold: AI is expected to help deliver the global development agenda.

That narrative matters because it shapes research framing, funding priorities, and policy expectations (Bush, 1945; Jasanoff, 2004; Stokes, 1997). Yet the apparent alignment it implies is usually inferred rather than demonstrated. Researchers and policymakers may converge on the same SDG labels while emphasising different problems, actors, timescales, and interventions. A paper and a policy document can both invoke SDG 3 (Good Health) or SDG 13 (Climate Action) without addressing the same substantive terrain. High-level thematic agreement can therefore mask deeper divergence. If so, current claims about "AI for SDGs" may overstate how far research effort is aligned with policy priorities, not because the narrative is necessarily false, but because the methods used to validate it remain too coarse.

The underlying problem is not new. Questions of research-policy divergence have been a central concern in policy studies and knowledge-utilisation scholarship for decades (Caplan, 1979; Guston, 2001; Haas, 1992; Weiss, 1979). What is new is not the embedding methodology itself, but its application to direct research–policy comparison within a shared SDG space. These tools allow broad discourse-level proximity to be assessed as an empirical property of text, rather than inferred only from anecdote, keyword coincidence, citation or mention counts, or aggregate coverage measures. In that sense, the aim of this paper is to map how closely academic AI-for-sustainability research and international institutional SDG policy discourse occupy nearby regions in semantic space.

Bibliometric work shows a persistent research-attention pattern, but rarely tests whether shared SDG labels imply shared semantic framing between research and policy discourse (reviewed in Section 2).

This study addresses that gap. Its central claim is straightforward: shared SDG labels are insufficient evidence of shared framing. I make three contributions. First, I advance AI–SDG mapping from Level 1 (lexical or bibliometric correspondence) to Level 2 (observed textual-semantic proximity in embedding space; Section 3.1). I treat this proximity as a register-sensitive proxy for framing, not as evidence of substantive alignment, using Sentence-BERT (Reimers & Gurevych, 2019) embeddings and a supervised logistic regression classifier trained on pooled SDG-labelled data from four independent annotated corpora. Second, I distinguish between a *coverage gap* (which SDGs each corpus emphasises) and a *semantic gap* (how differently the two corpora discuss the same SDG), and measure both across all 17 SDGs. Third, I test whether coverage and framing are related across the 17 SDGs, or whether SDGs that both corpora cover can still be discussed in very different ways.

The central research question is therefore: *To what extent do academic AI-for-sustainability research and international institutional SDG policy discourse differ in SDG coverage and semantic framing across the 17 SDGs, and how are these two dimensions related?*

In brief, the two corpora differ in both what they prioritise and how they frame those priorities, and the pattern of divergence is informative.

The remainder of this paper is structured as follows. Section 2 reviews the relevant literature on AI-SDG mapping, research-policy alignment, and semantic similarity methods. Section 3 describes the corpora, measurement instrument, and gap analysis procedures. Section 4 presents the empirical findings. Section 5 interprets the findings and discusses limitations. Section 6 concludes with implications and directions for future research.

2 Literature Review

2.1 AI and the SDGs: What Bibliometrics Shows

This section reviews past studies on AI and the SDGs. Most ask which goals draw the most research attention.

The best-known work, such as Vinuesa et al. (2020)'s assessment of AI's effects on the 169 SDG targets, rests on expert views or counts rather than on the texts themselves.

Broader SDG scientometric work has mapped the growth, institutional distribution, thematic structure, and interlinkages of MDG/SDG-related research across methods and time windows (Bautista-Puig et al., 2021; Cowls et al., 2021; Nedungadi et al., 2024; Singh et al., 2024). Bautista-Puig et al. (2021) chart the general SDG research landscape across higher-education and research institutions from 2000 to 2017 using a citation-based corpus of roughly 25,000 publications, finding SDG 3 (Good Health) by far the most addressed goal and SDG 7 and SDG 12 among the least. The AI-specific inventories converge on SDG 3 as dominant: Singh et al. (2024) map 20,511 AI publications (2001–2020) and rank SDG 3 and SDG 7 as the most applications-rich goals with SDG 17 the sparsest, while Cowls et al. (2021) survey 108 AI-for-social-good projects and report SDG 3 as the leader and SDGs 5, 16, and 17 each addressed by fewer than five. Nedungadi et al. (2024) apply BERTopic modelling to 1,288 big-data and AI publications (2013–2023) and likewise find SDG 3 most prominent, with SDG 9 and SDG 7 next. Across these studies, SDG 3 is the one robustly dominant goal, while the least-covered goals differ between the AI-specific inventories and the broader bibliometric picture — but none of them measures within-SDG semantic framing.

The methodological concern raised by Armitage et al. (2020) is important: SDG publication mapping depends on interpretive choices about what counts as SDG relevance, how target concepts are translated into queries or labels, and which bibliographic database is used. The embedding-based approach used here reduces some vocabulary-level brittleness of keyword dictionaries, but it remains a distributional measurement strategy whose results are conditional on corpus construction, centroid provenance, and model choice. Later benchmarking work reaches the same caution from a broader comparison of Boolean, machine-learning, and hybrid SDG-mapping systems: Kashnitsky et al. (2024) find that no single approach performs best across all validation datasets, and that no single "golden" SDG validation set exists.

Not only are SDG mapping systems methodologically contingent; the SDGs themselves are interdependent by design. Nilsson et al. (2016) observe that "the goals depend on each other" but that no operational framework specifies how, warning that treating targets in isolation "risk[s] perverse outcomes." Single-SDG assignment is therefore a simplifying assumption, not a reflection of how goals operate in practice.

One SDG breaks the interdependence pattern. Le Blanc (2015), in a network analysis of the proposed SDGs, treats SDG 17 (Partnerships for the Goals) as the dedicated goal for cross-cutting means of implementation — finance, technology transfer, capacity-building, and partnerships — and excludes it from cross-goal network mapping because its targets are implementation infrastructure rather than substantive domains: a distinction unique among the 17 goals. The Griggs et al. (2017) guide likewise describes SDG 17 as the main enabler of the whole framework. Unlike every other SDG, whose discourse centres on a shared substantive topic, SDG 17 policy discourse concerns coordinating and resourcing implementation itself — a conceptual space that technical AI research abstracts are not built to occupy.

Past work shows which SDGs attract AI and sustainability research, and how much that mapping depends on method and corpus. Whether research and policy texts for the same goals actually align is the subject of the next section.

2.2 Research-Policy Alignment in AI and Sustainability

This section asks whether AI research and sustainability policy line up, and why past work says they often do not.

A growing literature documents structural tensions between what AI research produces and what sustainability policy requires. Strauss et al. (2025) show that corporate AI research concentrates on pre-deployment technical problems, creating a gap between research priorities and the real-world deployment impacts that governance and policy must address. Sioumalas-Christodoulou and Tympas (2025) find that global AI metrics focus on technical and economic performance while policy discourse emphasises ethical and social dimensions aligned with SDG-related concerns. Toney-Wails et al. (2024), in the trustworthy-AI governance literature, provide a close methodological precedent, showing that policy and research communities can use shared trustworthy-AI terminology while differing in term salience, definitions, and focal concerns.

McCarthy et al. (2026) report a parallel finding in conservation science, where a multi-target SciBERT classifier applied to Ross Sea MPA research reveals that 59% of policy-prioritised research topics appear in fewer than 15 papers, demonstrating that research-policy coverage gaps are observable across domains.

Adjacent scientometric work has also used policy documents to measure research-policy connection, but at the level of mentions rather than semantic framing. Bornmann et al. (2016) examine climate-change publications in Altmetric-tracked policy documents and find that only 1.2% of 191,276 publications had at least one policy mention. Their study is useful here as a boundary marker: policy-document mentions can indicate one narrow form of policy uptake, but they do not show how research is discussed, interpreted, or framed inside policy discourse. The present study therefore addresses a different layer of the research-policy relationship: not whether policy documents cite particular papers, but whether research and policy-facing texts assigned to the same SDG occupy nearby regions in semantic space.

A related research-evaluation literature treats the research-policy relationship through the lens of societal impact measurement. Bornmann (2017) argues that measuring societal impact is harder than measuring scientific impact because the target of impact must first be specified: research may affect policymakers, business, culture, public health, or other social domains. Policy documents are therefore useful as one possible target-oriented source for measuring impact on policymaking, but this is a different task from semantic framing comparison. The present study does not ask whether research has achieved policy impact; it asks whether research and policy-facing texts occupy similar semantic regions when organised through shared SDG labels.

These findings are consistent with theoretical accounts of research-policy misalignment. Heink et al. (2015) identify credibility, relevance, and legitimacy as determinants of whether scientific knowledge informs policy; topical relevance alone is insufficient — relevance also depends on whether research results can shape actual decisions. In sustainability science, Cash et al. (2003) make a closely related argument: knowledge systems are more likely to support action when they produce information that is simultaneously salient, credible, and legitimate, and when they manage the boundary between knowledge and action through communication, translation, and mediation. The epistemic communities framework (Haas, 1992) provides a theoretical reason why shared framing matters for policy coordination under uncertainty: expert communities can influence policy by articulating cause–effect relationships, framing issues for collective debate, proposing policies, and identifying salient points for negotiation. However, Haas's concept is broader than semantic proximity alone, involving shared normative beliefs, causal beliefs, validity criteria, and a common policy enterprise. The present study therefore does not operationalise an epistemic community directly; it measures the weaker condition of observed semantic proximity between research and policy-facing discourse.

Cross (2013) strengthens this caution by arguing that epistemic communities should not be treated as simply present or absent: their influence varies with internal cohesion, professionalism, access to decision-makers, competition with other actors, and broader political context. This means that semantic proximity can at most indicate a possible precondition for knowledge transfer, not the existence, strength, or influence of an epistemic community.

The SDG framework is itself a politically negotiated instrument with documented limitations. Fukuda-Parr (2016) argues that global goals can have "distorting effects, redefining the meaning of development, and shaping policy by creating incentives to over-focus on target achievement." Bexell and Jönsson (2017) find that SDG summit documents remain "state-centric with great room for state sovereignty," avoid assigning causal responsibility for structural problems, and rely on "weak and unsystematic accountability."

A more fundamental tension exists within the framework itself. Hickel (2019) demonstrates that "Goal 8 violates the sustainability objectives of the SDGs" — 3% annual global growth is empirically incompatible with the resource-use and emissions targets in Goals 6, 12, 13, 14, and 15. The SDGs "lack theoretical grounding" and embody a contradiction between growth and ecological sustainability.

Heilinger et al. (2024) warns that the "sustainable AI" label can be used for greenwashing where technologies are framed as contributing to the SDGs without sufficient basis. Heras-Saizarbitoria et al. (2022), analysing 1,370 sustainability reports, find that most organisations' SDG engagement is superficial — "the SDGs only serve to add colour and fancy icons to reports" — suggesting that SDG-washing is a measurable empirical pattern. This supports the caution that reported alignment should not be conflated with substantive commitment.

Measuring semantic alignment with SDG centroids therefore uses a benchmark that is itself politically negotiated and internally inconsistent — not a neutral standard. Gibbons et al. (1994) distinguish Mode 1 knowledge production, where problems are set and solved within academic disciplinary interests, from Mode 2, where problems arise in contexts of application; to the extent that research follows Mode 1, its agendas may reflect internal disciplinary dynamics rather than external policy needs. This observation also operates alongside a further ambiguity: what it means to be "aligned" with a framework whose component goals are in tension with each other, and it suggests that divergence is not uniformly problematic — where research explores possibilities that policy discourse has not yet absorbed, distance may reflect productive exploration rather than failure of alignment.

The two-communities theory (Caplan, 1979) holds that researchers and policy makers live in separate worlds with different values, reward systems, and languages, so divergence is the default state of research-policy interaction rather than a failure. Taken together, this literature indicates that the divergence between AI research and sustainability policy is a substantive feature of the two knowledge systems rather than an artefact of measurement. Existing studies document the gap through coverage, citation, and impact-based methods, while theory explains it through legitimacy, expert communities, and the structural divide between academic and policy communities. None of these studies measures semantic framing within a single SDG. This study measures how close research and policy texts lie in semantic space; that closeness is a first step, not proof of alignment.

2.3 Semantic Methods: Rationale and Cautions

This section explains the method behind this study and why it needs care.

The use of Sentence-BERT (Reimers & Gurevych, 2019) and related transformer-based encoders for semantic comparison has been validated across several domains relevant to this study, but applying them to compare two distinct discourse corpora — research abstracts and policy documents — is a novel usage rather than a directly benchmarked one. This method follows the distributional hypothesis that meaning can be described through patterns of linguistic co-occurrence, a principle whose modern computational implementations trace back to Harris (1954)'s foundational observation that the distribution of a linguistic element is "the sum of all its environments." Sentence-BERT and related transformer-based encoders are therefore used here as pragmatic tools for large-scale semantic comparison, while the interpretation of distance remains descriptive rather than causal. Gjorgjevikj et al. (2025) benchmarked sentence encoders specifically for SDG association tasks, finding that general-purpose models from the sentence-transformers family perform competitively as baselines, and that domain-specific fine-tuning further improves predictive performance while reducing sensitivity to description length.

Rodriguez et al. (2023)'s conText framework provides a closer computational-social-science precedent for comparing meaning across contexts. Their embedding-regression approach estimates how focal terms are used differently across covariates such as party, time, or corpus, while explicitly treating embedding-based "meaning" as a thin distributional description rather than a causal model of human interpretation. This study does not adopt their word-level regression estimator, but it follows the same broad premise that semantic representations can be used descriptively to compare how different discourse communities frame shared terms or categories.

McCarthy et al. (2026) supply a yet closer empirical precedent for the present design. Applying a multi-target SciBERT classifier to Ross Sea MPA research and policy-prioritised topics, they find that 59% of policy-prioritised research topics appear in fewer than 15 papers, showing that research–policy coverage gaps are measurable with transformer encoders outside the AI–SDG setting. This study extends that logic from a single conservation domain to the full SDG landscape, and from coverage to within-SDG semantic framing.

An older corpus-linguistic tradition gives a further reason for caution when interpreting cross-corpus semantic distance. Biber's multidimensional analysis of spoken and written English showed that register differences are not simple binary contrasts, but continuous patterns of co-occurring linguistic features across genres (Biber, 1988). For the present study, this matters because research abstracts and policy documents may differ not only in substantive topic, but also in abstraction, modality, institutional stance, and rhetorical function. Embedding distance is therefore treated as a measure of observed textual-semantic proximity, not as a clean measure of topic alone.

Embedding models let us compare research and policy texts, but applying them across two corpora is new. They represent meaning through distribution, not cause, and cannot separate topic from register. The measures used here are descriptive: they report how close or distant the two corpora are in semantic space, not whether that closeness is real alignment.

2.4 Research Gap

To the best of my knowledge, the literature reviewed above has not produced: (i) a systematic comparison of how academic AI-for-sustainability research and institutional SDG policy discourse differ in SDG coverage and within-SDG semantic framing — prior work measures research coverage, citation patterns, or policy-document uptake, but does not compare the two corpora directly on these dimensions; or (ii) an examination of whether coverage gaps and semantic gaps are positively related, weakly related, or inversely related. These are the contributions of the present study, and they answer the central research question set out in the Introduction. Together, they provide a systematic framework for distinguishing research-policy coverage from research-policy semantic proximity within a shared SDG representation. This framework is deliberately diagnostic: it maps where observed divergence is visible, without classifying whether any given gap reflects productive or problematic disconnection — the latter would require qualitative follow-up beyond the scope of embedding analysis.

3 Methodology

3.1 A Three-Level Framework of Research-Policy Alignment

This study measures *topical overlap* between research and policy-facing discourse using semantic proximity in embedding space. I do not claim to measure substantive alignment, causal influence, or operational alignment — shared vocabulary does not guarantee shared meaning, but its absence makes meaningful dialogue structurally impossible. Nor should the measure be interpreted as a societal-impact metric: it captures observed textual-semantic distance between two corpora, not policy uptake (Bornmann et al., 2016) or shared causal beliefs (Haas, 1992). Embedding distances describe differences in textual context and usage, not direct effects on human understanding or policy uptake.

Following Biber and Conrad's distinction between register, genre, and style (Biber & Conrad, 2009), I treat the research-policy contrast primarily as a register contrast — the two corpora are produced in different institutional settings for different audiences and purposes. Cross-corpus register differences are discussed in Section 2.3 and bounded in Section 5.4.

I propose that research-policy alignment be understood as a three-level construct rather than a binary property of shared labels:

Level 1 — lexical or bibliometric correspondence. The dominant mode of existing SDG-mapping work, assigning texts to goals through keyword dictionaries, Boolean queries, or citation and mention counts (Armitage et al., 2020; Kashnitsky et al., 2024). Necessary but weak: shared vocabulary does not guarantee shared meaning (Bornmann et al., 2016), and keyword methods are brittle to vocabulary and database choices (Armitage et al., 2020).

Level 2 — observed textual-semantic proximity in embedding space. Sentence embeddings locate texts in a shared semantic space where proximity reflects distributional patterns of co-occurrence (Harris, 1954; Reimers & Gurevych, 2019), and encoder-based SDG association has been benchmarked as a competitive baseline (Gjorgjevikj et al., 2025). This level detects divergence that keyword methods miss, while remaining a proxy rather than a definitive conclusion: embedding distance mixes modality, abstraction, institutional stance, and theme (Biber, 1988), and the research-policy contrast is primarily a register contrast (Biber & Conrad, 2009). The present study is positioned at this level.

Level 3 — substantive comparison of implied problems, actors, and interventions. The ultimate object of alignment is whether the two discourses address the same problems, involve the same actors, and propose the same interventions (Cash et al., 2003; Guston, 2001; Haas, 1992). Because this requires interpreting causal beliefs, legitimacy, and institutional mechanism (Caplan, 1979; Haas, 1992), it lies beyond unsupervised text analysis and is left to qualitative follow-up.

Advancing from Level 1 to Level 2 is the study's concrete methodological contribution; Level 3 names the boundary it deliberately does not cross.

3.2 Research Corpus

This study uses only publicly available text data and codebase. It does not involve human participants or personal data. No ethical approval was required.

The research corpus consists of 3,105,144 English-language research abstracts retrieved from the OpenAlex API (Priem et al., 2022), covering publications between 2018 and 2025. Abstracts were retrieved by combining OpenAlex's native SDG work filter with four AI terms (`artificial intelligence`, `machine learning`, `deep learning`, `neural network`) — 68 queries (17 SDGs × 4 terms). Complete query parameters are documented in `code/0_fetch/fetch_openalex.py`. OpenAlex SDG metadata defines the candidate universe; the SDG assignments used in the analysis are produced by the independently validated supervised classifier (Section 3.5).

The choice of the four AI retrieval terms — `artificial intelligence`, `machine learning`, `deep learning`, `neural network` — was made on initial intuition rather than any systematic or validated basis, and was fixed before analysis began; it is therefore not defended or varied here. The corpus should be understood as conditional on this term selection, which may under- or over-represent subfields of AI-for-Sustainability research. The 2018 start date reflects the post-AlexNet (Krizhevsky et al., 2012) emergence of deep learning as the dominant paradigm and a reasonable research-uptake lag after the SDGs' adoption in 2015.

An important asymmetry follows: the research corpus is a tight intersection of AI and SDG content (AI ∩ SDG), while the policy corpus (Section 3.3) is a broader union covering institutional SDG discourse and AI governance. Between-SDG rankings should be interpreted as conditional on this asymmetric construction. Following Armitage et al. (2020), this retrieval universe is a method-dependent corpus boundary, not a neutral population of all SDG-relevant AI research. After deduplication and removal of records lacking usable abstracts, 3,105,144 unique papers were retained. No balancing by SDG or citation count is applied — trimming would distort the coverage profile the study aims to measure.

3.3 Policy Corpus

The policy corpus contains 43,147 text segments from three collections, representing 2,455 unique source documents. It should be read as mixed *international institutional SDG* *policy discourse* rather than the full universe of AI sustainability policy:

1. **Curated AI and SDG policy documents** (15,639 segments, 95 documents): UN, OECD, EU, UK, and AU AI governance frameworks; IPCC assessment reports; SDSN sustainable development reports; and national AI strategies. These are AI-governance instruments by construction, forming the innovation-facing slice of the corpus.
2. **SDGi Voluntary National Review and Voluntary Local Review corpus** (SDGi, 2024) (31,971 segments, 313 documents): National and local self-assessments submitted to the UN High-Level Political Forum. The creators describe it as the most comprehensive multilingual collection of SDG-labelled texts to date, spanning all 17 goals and centred on SDG implementation rather than any single sector.
3. **UN General Debate Corpus** (Harvard Dataverse, 2024) (6,472 segments, 2,048 documents): SDG-relevant passages from General Debate speeches, sessions 70–80 (2015–2024). Its creators analyse SDG coverage directly and report climate action (SDG 13), peace and governance (SDG 16), and partnerships (SDG 17) among the most prominent themes — a climate–governance–partnership orientation the analysis below also observes.

Each source document was segmented using the transformer model's own tokenizer: NLTK sentence-boundary detection (`sent_tokenize`) is followed by greedy accumulation of sentences up to a token budget of `max_seq_length - 10` (margin 10, empirically verified to reduce truncation to near zero). Segments falling below 20 words (`MIN_SEGMENT_WORDS`) are discarded; a single sentence exceeding the token budget is kept whole because mid-sentence splitting is not permissible. After deduplication by exact text match, 43,147 unique segments remained. Segment identifiers encode source document identity for document-level weighting.

Source heterogeneity note. The three collections differ in genre and scope — the curated subset is AI-governance-focused, while SDGi and UNGDC reflect broader institutional SDG discourse. The full-corpus policy distribution should be read accordingly; the curated sub-corpus (95 documents) is the more relevant reference for AI-governance-specific interpretation. Appendix A.2 reports a source-family sensitivity check.

3.4 Embedding Model and Normalisation

All texts were encoded using `all-mpnet-base-v2` (Sentence-Transformers, 2021) from the sentence-transformers library (Reimers & Gurevych, 2019), producing 768-dimensional dense vector representations. This pre-trained model achieves competitive performance on semantic textual similarity benchmarks and requires no fine-tuning on the data under analysis — the embedding space is a stable pre-existing coordinate system. All embeddings were L2-normalised to unit vectors so that cosine similarity between any two vectors equals their dot product.

During pipeline development, a truncation issue was discovered and corrected: texts exceeding the encoder's maximum sequence length were being silently truncated before embedding, discarding tail content. The fix uses the model's own tokenizer during segmentation (Section 3.3) to enforce a token budget of `max_seq_length - 10`, verified by the `verify_truncation_rate` utility to produce near-zero post-segmentation truncation. The full corpus was re-embedded after the fix; all reported results use post-fix embeddings.

3.5 Supervised Reference Classifier

The canonical instrument is a supervised multinomial logistic regression (LR) classifier trained on pooled single-label reference data from four independent annotated corpora:

- **OSDG Community Dataset** (Pukelis et al., 2022) (30,534 texts, SDGs 1–16): a crowdsourced citizen-science collection of excerpts from reports, policy documents, and abstracts, each validated by multiple volunteers.
- **IISD SDG Knowledge Hub** (Wulff et al., 2023) (2,221 texts, SDGs 1–17): IISD journalism and policy commentary on 2030 Agenda implementation.
- **SDGi Corpus** (SDGi, 2024) (5,233 texts, SDGs 1–17): government Voluntary National and Local Reviews submitted to the UN.
- **Aurora dataset** (Vanderfeesten et al., 2020) (5,619 expert-validated texts, SDGs 1–17): academic abstracts validated by domain researchers.

All four corpora supply human- or expert-validated labels rather than machine-generated ones, and they differ in provenance. OSDG and SDGi are policy-adjacent: OSDG volunteers confirm a suggested label across multiple annotators, and SDGi compiles government Voluntary National and Local Reviews submitted to the UN. The SDG Knowledge Hub is journalism and reporting; Aurora is academic, expert-validated text. The mix broadens the label base, but OSDG alone contributes about 70% of all labelled texts and draws on UN-adjacent documents, so the reference data remain weighted toward policy vocabulary. No high-quality graded relevance sets exist for SDG classification, making this combined, multi-source collection the best available resource (Kashnitsky et al., 2024).

The pooled reference data (n=52,779 after deduplication and preprocessing) were split into training and held-out test sets (n=52,779 train, n=9,734 test) by stratified document-grouped splitting: all texts from the same source document are kept in the same fold, enforcing strict separation between training and evaluation. The champion LR (C=10.0, L2 penalty, class_weight=None, solver=lbfgs) was selected by 5-fold GroupKFold cross-validation on the training pool; the full grid-search protocol, hyperparameter rationale, exclusion of alternative model families (RF, XGB), and the comparison with the MLP robustness check are reported in Appendix D. The main text states plainly that LR is the canonical method; the MLP is a robustness check, not an alternative champion.

The validation task evaluates whether the classifier assigns held-out test examples to the correct SDG; it does not evaluate a rejection option for non-SDG texts. This is acceptable because the research corpus is first restricted to an OpenAlex AI-and-SDG universe — the classifier then answers a relative question: among the 17 SDGs, which is this text closest to?

3.6 Scoring and Assignment

A text is assigned to the SDG with the highest predicted probability from the trained LR classifier (argmax over 17 classes). Every assignment is traceable to 768 signed logistic-regression coefficients per SDG class — a transparent decision boundary with no hidden layers, no dropout, and no random seed sensitivity. Macro-F1, the unweighted mean of per-SDG scores, is the primary validation metric; the scores are reported in Section 4.1 alongside SDG-specific concerns that inform the results below.

LR argmax assigns each text to the SDG whose decision boundary it falls on the positive side of. The method is simple, fast, and avoids the hidden-layer opacity of non-linear classifiers. I chose it deliberately: the study needs a transparent tool to compare corpora, not a classifier built to label single documents. LR additionally provides calibrated per-class probability outputs, which can serve as a soft-assignment comparison against the hard argmax assignment (Appendix B.2); the canonical analysis uses hard argmax for direct interpretability.

Additional diagnostic checks — within-corpus diagnostics, policy source-family sensitivity, and an SDG 4 lexical audit — are reported in Appendix B. The canonical analysis uses hard LR argmax assignment as a transparent and reproducible anchoring procedure.

3.7 Coverage Gap Analysis

For each corpus item, the predicted SDG is the LR argmax (hard assignment, Section 3.6). The coverage profile is the proportion of items assigned to each SDG.

The policy corpus is not a flat set of documents: its 43,147 segments are grouped into 2,455 source documents, and those documents vary enormously in length (SDGi: 31,971 segments across 313 documents; curated AI/SDG: 15,639 across 95; UNGDC: 6,472 across 2,048). A segment-weighted profile would let a handful of long reports dominate the count, so I instead compute a *document-weighted* profile in which every source document counts equally regardless of length. Concretely: (1) the segment score vectors of each document are averaged into one document-level vector; (2) that vector is hard-assigned to its top

SDG; (3) the coverage profile is the share of documents — not segments — assigned to each SDG. The research corpus uses the same logic at the paper level, where each of the 3,105,144 papers is already a single unit. Document-weighted profiles are the canonical representation; segment-weighted profiles are retained as a diagnostic comparison.

The coverage gap for SDG j is defined as:

$$\text{CoverageGap}_j = \left|\text{ResearchCoverage}_j - \text{PolicyCoverage}_j^{\text{docwt}}\right|$$

The signed version of the same quantity is:

$$\text{SignedCoverageGap}_j = \text{ResearchCoverage}_j - \text{PolicyCoverage}_j^{\text{docwt}}$$

A positive value means research dominates SDG j; a negative value means policy dominates. These two definitions, together with research coverage and policy coverage, are the four predictors reported in Table 2.

3.8 Semantic Gap Analysis

The semantic gap measures within-SDG divergence: given that both corpora have texts assigned to SDG j, how differently do they discuss it? For each SDG j, I collect all assigned research papers and all assigned policy segments (capped at 50 per source document to prevent single-document dominance — SDSN contributes up to 3,179 segments otherwise), compute the L2-normalised mean embedding of each cluster ($\mathbf{c}_j^{\text{res}}$, $\mathbf{c}_j^{\text{pol}}$), and define:

$$\text{SemanticGap}_j = 1 - (\mathbf{c}_j^{\text{res}} \cdot \mathbf{c}_j^{\text{pol}})$$

A gap of 0 indicates perfectly overlapping semantic directions; a gap approaching 1 indicates orthogonal framing.

The policy corpus is represented as short segments rather than whole documents, so a per-SDG policy sub-centroid is a mean over all assigned segments. Without limitation, a handful of very long documents would dominate the cluster: within the curated AI/SDG collection, SDSN sustainable development reports alone contribute up to 3,179 segments, and the SDGi Voluntary National and Local Review reports together span 31,971. I therefore apply a *per-document segment cap*: for each source document, at most K of its segments are randomly sampled (seeded, hence reproducible) before the sub-centroid is built, so that no single document can outweigh the rest. Random rather than positional sampling (e.g. taking the first K segments) avoids biasing the centroid toward a document's opening or closing passages. The canonical cap is $K = 50$ (≈7,500 words), a conservative choice well above the median of roughly 14 segments per document. The cap choice was checked by repeating the computation at an alternative cap of 20 segments per document and with no cap at all; the SDG gap rankings are near-identical to the canonical cap-50 results (rank correlation $\rho = 0.966$; Table 3, Section 4.6), so the findings are robust whether or not the cap is applied.

The semantic gap is not a clean measure of substantive disagreement. Following Biber and Conrad's distinction between register, genre, and style (Biber & Conrad, 2009), I treat the research–policy contrast primarily as a register contrast: the two corpora differ in situation of use, audience, and communicative purpose, not merely in topic. Embedding distance therefore mixes a genuine within-SDG substantive divergence (the two corpora discuss the same SDG differently) with register divergence (research abstracts and policy documents differ in abstraction, modality, and institutional stance) (Biber, 1988). The raw centroid distance is retained as the primary measure because it directly corresponds to this observed framing difference. Separating the substantive component from the register component is an open problem. The adjusted estimates reported in Table 3 bound the register-sensitive component but do not replace the raw gap.

3.9 Coverage–Semantic Interaction

The first hypothesis (H1) decomposes into four named sub-hypotheses, each correlating a distinct coverage predictor with the within-SDG semantic gap across the 17 SDGs: H1a, coverage gap $\leftrightarrow$ semantic gap; H1b, research–policy dominance $\leftrightarrow$ semantic gap; H1c, research coverage $\leftrightarrow$ semantic gap; and H1d, policy coverage $\leftrightarrow$ semantic gap. H1a is the primary, motivating hypothesis: the absolute research–policy coverage gap is expected to predict semantic divergence. H1b is its directional complement, testing whether the signed balance of research and policy attention (which side dominates a goal) also predicts divergence. I compute Pearson r and Spearman ρ between each coverage predictor and semantic gap scores across all 17 SDGs, with sensitivity analyses excluding SDG 4 (see Section 3.10). Given $n = 17$, the tests have limited power to detect anything but a large linear effect; this is a scope condition.

3.10 Key Assumptions and Mitigations

This subsection records the design-level assumptions that shape the comparison before any measurement takes place. Two such choices are methodological decisions rather than empirical findings: how text units are handled across the two corpora, and how a known limitation of the SDG 4 reference labels is treated.

Text-unit handling is a design choice, not a result. Research papers are single abstract-length texts (mean $\sim$200 words); policy items are 150–300-word segments from documents of highly variable length. Document-weighting and per-document segment caps are adopted up front to keep the two corpora comparable; they are fixed procedures, not findings derived from the data.

Labelled-data limitation on SDG 4. The research coverage assigns 14.6% of papers to SDG 4 (Quality Education, see later Section 4.3), flagged as artefact-affected because ML papers share core vocabulary (“learning”, “training”, “model”) with the OSDG education corpus. Appendix A.4 confirms that SDG 4-assigned texts contain education-specific terms more often than matched non-SDG4 samples, but co-occur with ML-learning vocabulary. The result is retained as a learning-related SDG 4 assignment rather than a clean education estimate. Sensitivity analyses excluding SDG 4 are provided for all correlation results. This is a limitation of the reference labels, carried through to the results and mitigated there by artefact-flagged sensitivity analyses, not resolved at the design stage.

4 Results

This section reports what the instrument measured: the validation scores, the coverage and semantic gaps, the H1 correlation tests, and the robustness of those rankings. It describes the findings; it does not interpret them. Interpretation, limits, and implications follow in the Discussion (Section 5).

4.1 Supervised Reference Classifier Validation

The LR classifier was evaluated on the held-out test set (n=9,734, document-grouped to prevent source leakage). It reaches macro-F1 = 0.823 and micro-F1 = 0.8369, far above the 1/17 random baseline (a 17-class random guess scores $1/17 \approx 0.059$). Per-class F1 values from the test set are reported in Table 1.

The LR achieves consistently strong per-SDG performance: SDG 3 (F1 = 0.910), SDG 14 (0.906), and SDG 4 (0.889) are the most confidently classified (each above 0.89). The weakest classes are SDG 10 (Reduced Inequalities, F1 = 0.637) and SDG 8 (Decent Work, F1 = 0.678), consistent with their known semantic breadth and the overlapping cluster structure of adjacent goals (see centroid-similarity heatmap, Figure 5). Most goals score between 0.76 and 0.89, indicating reliable discrimination across the 17-class space. SDG 17 (F1 = 0.797) is no longer the weakest link — the supervised decision boundary separates partnership discourse more cleanly than the centroid average did — confirming that the goal’s broad mandate resists compact semantic definition regardless of method.

The MLP robustness check (Appendix D) achieves a comparable test macro-F1 (MLP: 0.836; LR: 0.823), confirming that architecture choice is not the binding constraint on classification performance. The gap between LR and MLP is modest and does not justify the complexity trade-off — LR provides 768 signed coefficients per class, each directly

Table 1. LR test-set F1 scores (n=9,734).

SDG	LR test F1
SDG 1 (No Poverty)	0.758
SDG 2 (Zero Hunger)	0.829
SDG 3 (Good Health)	0.910
SDG 4 (Quality Education)	0.889
SDG 5 (Gender Equality)	0.883
SDG 6 (Clean Water)	0.878
SDG 7 (Clean Energy)	0.874
SDG 8 (Decent Work)	0.678
SDG 9 (Industry & Infra.)	0.780
SDG 10 (Reduced Inequalities)	0.637
SDG 11 (Sustainable Cities)	0.789
SDG 12 (Responsible Cons.)	0.786
SDG 13 (Climate Action)	0.843
SDG 14 (Life Below Water)	0.906
SDG 15 (Life on Land)	0.870
SDG 16 (Peace & Justice)	0.886
SDG 17 (Partnerships)	0.797
Macro-F1 (SDGs 1–17)	**0.823**

Notes: LR test F1 on the held-out pooled test set (all reference sources). The table shows the canonical 17-way classifier used throughout the main analysis.

attributable to a semantic dimension, while MLP's hidden layers diffuse this information across non-linear transformations.

In summary, the LR classifier separates the 17 SDGs with consistent and interpretable performance. The weakest classes — SDG 10 and SDG 8 — are known to sit in dense centroid clusters, and SDG 17, while no longer the lowest-F1 goal in the LR test set, remains the most method-sensitive SDG in the cross-sensitivity analysis. These scores mark a ceiling for the embedding-plus-linear-classifier approach; MLP results confirm that the ceiling is not an artefact of linear separability assumptions.

4.2 Scoring Research and Policy Corpora on the Measurement Instrument

This subsection reports how the two corpora actually score against the measurement instrument, and where the scoring asymmetries come from. It is the empirical companion to the instrument-construction caveats in Section 3.5, and the design choices that frame them are recorded in Section 3.10.

What "score" means. "Score" is each text's highest LR predicted probability — the maximum over 17 SDG classes (the same argmax value that drives hard assignment, Section 3.6).

Corpus-level score asymmetry. Policy segments score systematically higher against the LR classifier than research papers do (mean 0.761 vs 0.656, a 0.105 gap). The gap has two causes the data cannot separate: (1) a genre artefact, because the classifier is trained on largely policy-adjacent reference corpora (Section 3.5), or (2) a substantive difference, because policy texts address SDG implementation more directly than research abstracts. Either way it does not threaten the findings: coverage rests on hard assignment — assigning each text to its highest-scoring SDG — so only the within-corpus ranking matters, not the absolute score level. Additionally, a structural check confirms that the asymmetry is not an SDGi circularity effect: the training set and the scored policy corpus are disjoint by construction (Appendix A.3), so the score gap cannot be driven by the classifier recognising its own training texts in a held-out corpus.

Read plainly, the gap means one thing. If it says anything, it is only that the LR classifier assigns higher probabilities to policy texts than to research texts. The magnitude is not itself an argument — it varies by goal (per-SDG policy–research top-score gap: min 0.060, max 0.287, mean 0.186, SD 0.069; positive for all 17 goals), but the direction, not the size, is the point. Because hard assignment uses only the within-corpus ranking, this directional gap does not reorder goals.

A rough view of the asymmetry. The combined PCA landscape (Figure 1) shows the policy cloud hugging the area around the 17 reference-classifier centroids (the per-SDG mean training embedding) more tightly than the research cloud. Research-paper and policy-text embeddings are combined into one corpus and projected onto the first two principal components; the 17 SDG centroids are overlaid with the same transform, alongside the full-corpus research (circle) and policy (square) centroids. The projection is descriptive only: the first two components explain just 9.1% and 3.7% of variance, so it is a coarse map of the 768-dimensional space, not a geometric measurement. All reported cosine distances remain in that original space (Section 4.4).

Summary. The policy–research score gap is real and uniform in direction (policy always receives higher LR probabilities) but its size varies by goal and its cause is unidentified. It is a measurement artefact to hold constant, not a finding about AI-for-sustainability content: hard assignment isolates the within-corpus ranking the analysis actually uses, and the structural disjointness check (Appendix A.3) confirms it is not an SDGi circularity effect. The reference data's policy-adjacency is a caution about reading proximity as alignment, not a defect of the method; within-corpus diagnostics in Appendix B show the same fuzz at corpus level.

4.3 Coverage Gap: What SDGs Each Corpus Prioritises

Figure 2 presents the document-weighted policy coverage profile alongside the research coverage profile. The two profiles are markedly different.

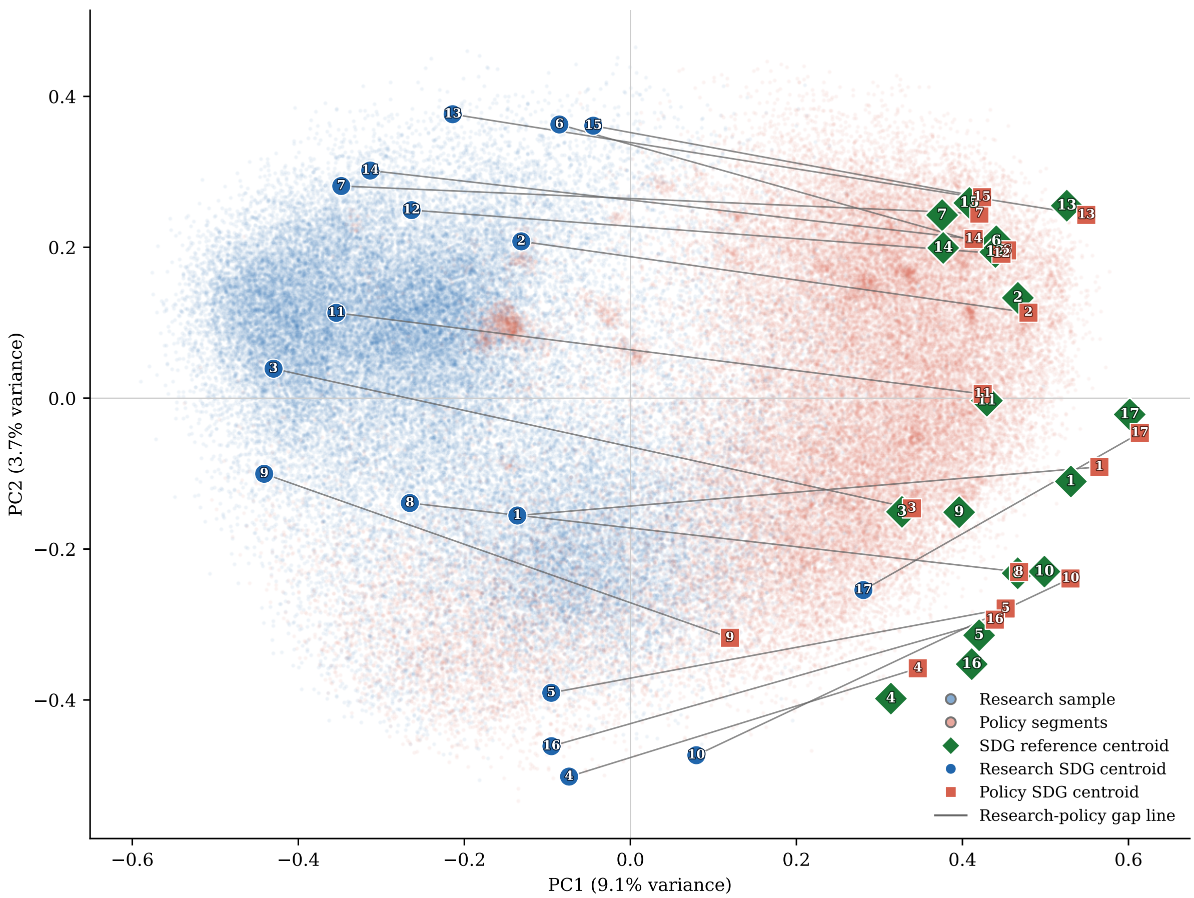

Figure 1. Figure 1. PCA semantic landscape of the combined research–policy embedding space. Descriptively two-dimensional only; formal semantic distances are computed in the original embedding space.

Policy corpus. Three SDGs dominate the document-weighted profile of the full institutional policy corpus: SDG 16 (Peace, Justice and Strong Institutions, 38.9%), SDG 13 (Climate Action, 20.2%), and SDG 17 (Partnerships for the Goals, 19.0%), accounting for 78.1% of policy documents. This ordering is consistent with the literature review's account of SDG 17 as the cross-cutting means-of-implementation goal whose discourse concerns coordination and resourcing rather than a substantive topic (Griggs et al., 2017; Le Blanc, 2015); the climate–governance emphasis also matches the SDGi and UNGDC policy components described in Section 3.3. The SDG 16 lead is notable: under the supervised classifier, governance and justice discourse emerges as the most prevalent policy theme, ahead of both climate and partnerships. Appendix A.2 shows that this full-corpus profile is source-family sensitive: the curated AI/SDG sub-corpus is much more innovation-heavy, while the broader SDGi and UNGDC components are more climate-, partnerships-, and governance-oriented.

Research corpus. The research profile is broader, but it still concentrates strongly on a smaller set of SDGs. The largest research shares are SDG 3 (Good Health, 25.3%),

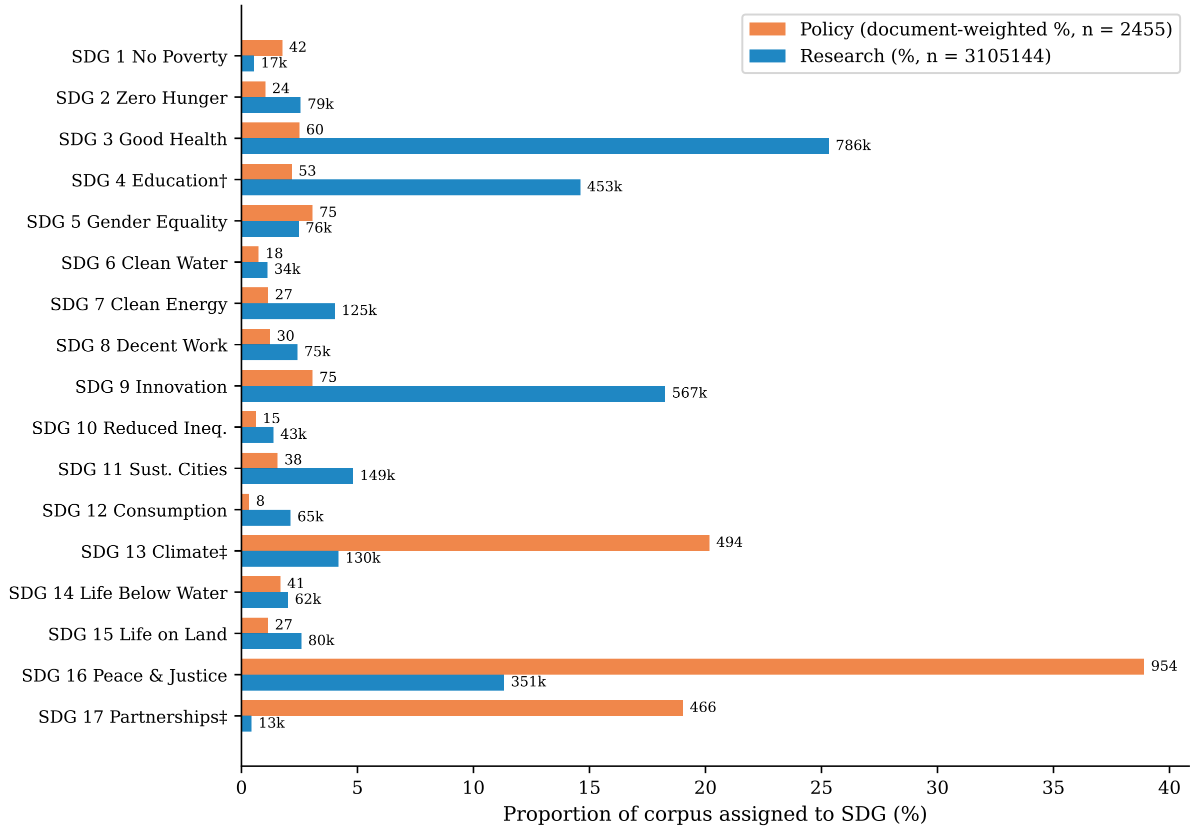

Figure 2. Coverage profiles by SDG.

SDG 9 (Innovation, 18.3%), and SDG 4 (Quality Education, 14.6%), followed by SDG 16 and SDG 11 at 11.3% and 4.8% respectively. The SDG 3 lead corroborates the literature review, where SDG 3 is the one goal every major AI and sustainability bibliometric study finds dominant — Bautista-Puig et al. (2021), Singh et al. (2024), Cowls et al. (2021), and Nedungadi et al. (2024) all rank it first, with SDG 9 and SDG 7 typically next. The SDG 4 share is the one point of disagreement: prior inventories do not rank SDG 4 among the leaders, so its third-place finish here is the same result flagged as likely inflated by machine-learning vocabulary (Section 3.10); read cautiously, the research profile agrees with the literature on SDG 3 and SDG 9 but over-states SDG 4. Even so, the research corpus clearly does not mirror the policy-side dominance of SDGs 16, 13, and 17.

Largest coverage mismatches. The five largest absolute coverage gaps are SDG 16 (0.276), SDG 3 (0.228), SDG 17 (0.186), SDG 13 (0.160), and SDG 9 (0.152). SDG 16 is now the most asymmetric goal — policy discourse strongly emphasises governance and institutions while research barely touches it — followed by the familiar research-over-policy dominance in health (SDG 3).

Taken together, three observations support reading the coverage profiles as a real signal rather than a measurement artefact. First, the instrument reaches macro-F1 = 0.823 on the held-out benchmark (Section 4.1), far above the 1/17 random baseline, so assignments are not random. Second, the research-side ranking reproduced here — SDG 3 dominant, with SDG 9 and SDG 7 next — matches the ordering reported independently by four prior bibliometric studies of AI and sustainability research Bautista-Puig et al. (2021), Singh et al. (2024), Cowls et al. (2021), Nedungadi et al. (2024) (Section 2), which used keyword and citation methods rather than embeddings. Third, this is all while the classifier is trained on reference corpora that are themselves policy-adjacent (Section 3.5). To summarize, that the research-covered SDG ranking nonetheless converges with independent, methodologically unrelated literature is the strongest available evidence that the classifier anchors a meaningful semantic coordinate system, not a policy-specific artefact.

4.4 Semantic Gap: How Differently Research and Policy Discuss Each SDG

Figure 3 shows the within-SDG semantic gap for all 17 SDGs. The gap ranges from 0.229 (SDG 15) to 0.478 (SDG 13), a span of 0.261 cosine units. A large gap means research and policy use different words and concepts for the same goal; a small gap means they frame it the same way. The largest gap is about 4.5 times the policy–research score gap (0.105, Section 4.2). SDG 13 has the largest gap, followed closely by SDG 3 and SDG 11. The ranking is stable when the segment cap is 20 per document or removed (rank correlation $\rho = 0.966$; Table 3), so the result does not depend on the sampling parameter.

Largest semantic gaps. SDG 13 (Climate Action, 0.478), SDG 3 (Good Health, 0.459), SDG 11 (Sustainable Cities, 0.456), SDG 1 (No Poverty, 0.443), and SDG 7 (Clean Energy, 0.433) show the widest gaps. The top-gap SDGs are a mix of climate, health, and infrastructure goals where research and policy discourse diverge most sharply. This ordering is stable across the segment-cap sensitivity check (Table 3). Notably, SDG 17 (Partnerships, 0.217) now shows the smallest semantic gap — the supervised decision boundary separates partnership discourse more cleanly than the old centroid method, moving it from the most to the least divergent goal.

Smallest semantic gaps. SDG 17 (0.217), SDG 15 (Life on Land, 0.229), SDG 4 (Quality Education, 0.264), SDG 6 (Clean Water, 0.283), and SDG 9 (Industry and Infrastructure, 0.283) show the narrowest gaps, meaning research and policy frame these goals in close terms. A narrow gap does not mean balanced attention: SDG 9 is still research-dominant, SDG 17 is policy-dominant, and SDG 4 carries the learning-vocabulary concern (Appendix A.4). Appendix B.1 shows example texts for high- and low-gap cases.

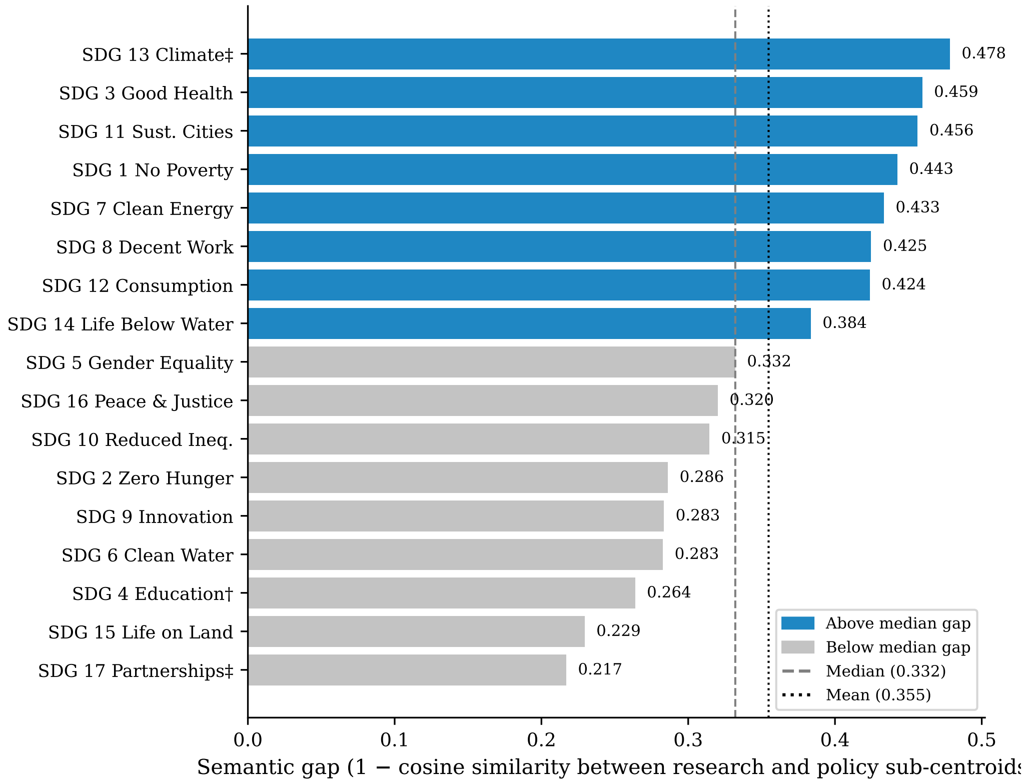

Figure 3. Within-SDG semantic gap by SDG, sorted descending. Dashed line = median (0.332); dotted line = mean (0.355). Dark bars exceed the median; light bars fall below it.

4.5 Coverage–Semantic Interaction: Two Distinct Dimensions of Observed Divergence

Table 2 reports the H1a–H1d correlation tests between coverage measures and semantic gap scores across the SDGs.

The four predictors are research coverage, policy coverage, the absolute research–policy coverage gap, and the signed research–policy dominance. Each panel shows the result across all 17 SDGs together with tests dropping SDG 4 and SDG 17.

Table 2. H1a–H1d correlation tests between coverage predictors and the within-SDG semantic gap.

Config	Pearson r	p	Spearman ρ	p
H1a. Coverage gap ↔ Semantic gap				
Primary observed SDGs (n=17)	-0.066	0.801	0.103	0.694
Excluding SDG 4 (n=16)	-0.030	0.911	0.191	0.478
Excluding SDG 17 (n=16)	0.070	0.797	0.259	0.333
H1b. Policy–research dominance ↔ Semantic gap				
Primary observed SDGs (n=17)	0.123	0.639	0.076	0.772
Excluding SDG 4 (n=16)	0.209	0.437	0.182	0.499
Excluding SDG 17 (n=16)	-0.048	0.861	-0.088	0.745
H1c. Research coverage ↔ Semantic gap				
Primary observed SDGs (n=17)	0.062	0.814	0.218	0.400
Excluding SDG 4 (n=16)	0.159	0.555	0.350	0.184
Excluding SDG 17 (n=16)	-0.020	0.941	0.062	0.820
H1d. Policy coverage ↔ Semantic gap				
Primary observed SDGs (n=17)	-0.099	0.707	0.092	0.725
Excluding SDG 4 (n=16)	-0.128	0.635	0.156	0.564
Excluding SDG 17 (n=16)	0.039	0.886	0.243	0.364

H1a does not hold in the canonical specification: the absolute coverage gap and semantic gap are near-uncorrelated ($r = -0.066$, $p = 0.801$; $n = 17$). H1c likewise does not hold: research coverage and semantic gap are uncorrelated ($r = 0.062$, $p = 0.814$). Two exploratory tests — H1d, policy coverage ($r = -0.099$, $p = 0.707$), and H1b, signed dominance ($r = 0.123$, $p = 0.639$) — are also near-zero. Dropping SDG 4 or SDG 17 leaves all associations unchanged (Table 2), and all four tests fail under every alternative assignment method, policy source family, and segment cap covered in the cross-sensitivity check (Table 3). The four predictors across eight configurations give 32 tests; none reaches significance.

Figure 4 shows each SDG. The most striking pattern is the absence of a pattern: SDGs with large coverage gaps span the full range of semantic gaps (e.g. SDG 16 has the largest coverage gap but a below-median semantic gap, while SDG 3 has a large coverage gap and the second-largest semantic gap). Research attention coexists with both low- and high-gap cases (e.g. SDG 9 high-coverage low-gap, SDG 3 high-coverage high-gap), which is why H1c is near zero. No SDG drives a positive association; the null result is consistent across all configurations.

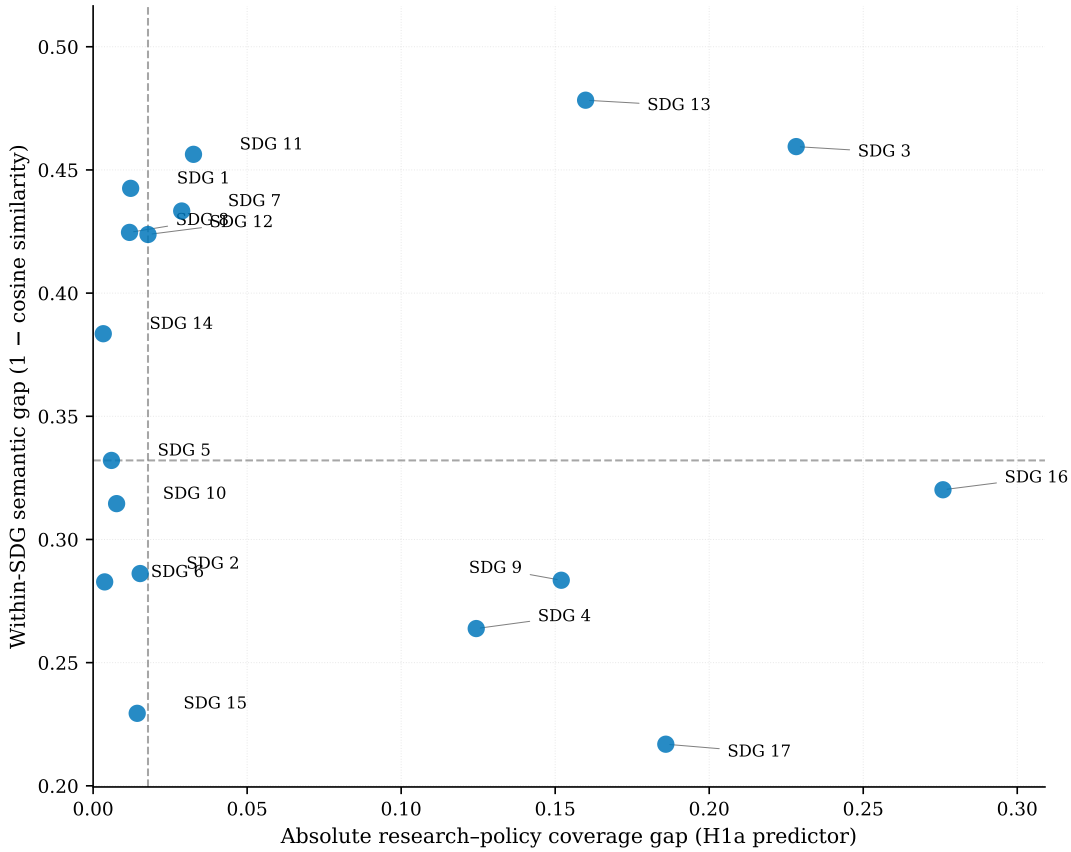

Figure 4. Coverage gap vs semantic gap, one point per SDG.

4.6 Cross-Sensitivity Robustness of the Gap Rankings

The semantic gap rankings discussed below are not equally stable across all measurement choices. Table 3 shows each SDG's gap rank (1 = largest gap) under the canonical configuration (LR, full policy corpus, cap 50) and eight alternatives spanning three axes: assignment method (zero-shot nearest-centroid, MLP), policy source family (curated AI/SDG, SDGi, UNGDC), and segment cap (20, none).

Three tiers emerge. First, SDG 17 is the most method-sensitive SDG: it sits at rank 17 (smallest gap) under both LR and MLP but rises to rank 7 under zero-shot assignment. SDG 13 and SDG 11 are the most robustly high-gap goals, staying in the top four under all three assignment methods (LR ranks 1, 3; zero-shot ranks 4, 2; MLP ranks 3, 4). SDG 3 also stays high across methods (LR 2, zero-shot 5, MLP 6). Second, several SDGs shift dramatically by assignment method alone: SDG 12 jumps from rank 7 under LR to rank 1 under MLP; SDG 5 moves from rank 9 under LR to rank 16 under zero-shot; SDG 8 moves from rank 6 to rank 15. Third, across policy-source and cap configurations, SDG 11 stays in the top five under every sub-corpus and cap; SDGs 13 and 3 also stay consistently high (ranges 1–7 and 1–10 respectively). SDG 4 is the most stable low-gap goal, staying in ranks 14–15 across all nine configurations. The discussion limits its claims to findings that survive this filtering.

Table 3. Cross-sensitivity robustness of within-SDG semantic gap rankings.

SDG	Assignment method			Policy source				Segment cap	
SDG	LR	Zero-shot	MLP	Full	Curated	SDGi	UNGDC	20	None
SDG 1	4	8	2	4	3	5	6	4	4
SDG 2	12	10	9	12	11	13	12	13	12
SDG 3	2	5	6	2	10	2	1	2	6
SDG 4	15	14	15	15	15	14	15	15	15
SDG 5	9	16	8	9	12	11	10	10	9
SDG 6	14	6	13	14	9	15	14	14	13
SDG 7	5	1	5	5	1	8	5	5	3
SDG 8	6	15	10	6	6	6	4	7	5
SDG 9	13	17	14	13	14	1	11	11	14
SDG 10	11	13	12	11	7	12	13	12	10
SDG 11	3	2	4	3	5	3	2	3	2
SDG 12	7	9	1	7	2	7	3	6	7
SDG 13	1	4	3	1	4	4	7	1	1
SDG 14	8	3	7	8	8	9	9	8	8
SDG 15	16	11	17	16	13	16	16	16	16
SDG 16	10	12	11	10	16	10	8	9	11
SDG 17	17	7	16	17	17	17	17	17	17

Notes: Each cell reports the gap rank (1 = largest gap, 17 = smallest gap) under each measurement configuration. The LR column is the canonical assignment method (cap 50 segment cap, full policy corpus); the Zero-shot column uses nearest-centroid assignment; the MLP column uses the 4-layer/384-hidden champion retrained on the full training pool. Segment-cap and policy-source columns are LR-based unless noted.

Caution: the score profile is source-sensitive. The coverage and gap profiles depend on how the corpora are composed. The policy source-family split shifts the full-corpus profile: the curated AI/SDG sub-corpus is innovation-heavy (SDG 9 leads), while SDGi and UNGDC are climate-, partnerships-, and governance-oriented (SDG 13, 17, 16 lead) (Tables 4, 5, Appendix A.2). The cross-sensitivity table (Table 3) reports gap ranks under all three sensitivity axes. The full-profile figures are conditional on the combined source mix; the per-family and per-method decompositions are the uncertainty bounds. The analysis uses all policy sources and the combined reference sources – the most data available – so the reader should hold these sensitivities in view. They are why the cross-sensitivity check bounds which findings are stable enough to interpret before any claim in Section 5.

5 Discussion

This section interprets the findings reported above. It moves from what the data show to what they mean: the structure behind the two divergence dimensions, the robust patterns, and the implications and limits of the study. No new results are introduced here.

This dissertation makes two contributions, and they carry different weight. The first is a measurement framework: a way to separate coverage from framing, and to compare research and policy text directly in the same embedding space. This framework holds up. It survives changes of assignment method, policy source family, and segment cap, as the checks in Section 4.6 and the appendices show. The second is a set of findings about how AI research and SDG policy actually relate: which goals each side covers, how far apart their language sits within a goal, and whether the two move together across goals. These findings are weaker. The relationship between coverage gaps and semantic gaps reverses sign under some configurations (Table 2), and SDG 17's position depends heavily on the assignment method used (Table 3). The framework should be read as the dissertation's main contribution. The findings about alignment should be read as a first application of it, worth checking again with a larger and more diverse policy corpus, not as a settled result.

5.1 Two Distinct Dimensions of Observed Divergence

The headline result of this study is simple: shared SDG labels are insufficient evidence of shared framing. The study separates two divergences — a coverage gap (how differently research and policy prioritise each SDG) and a within-SDG semantic gap (how differently they frame it) — and asks whether the first predicts the second. The primary test (H1a) finds no relationship in the canonical specification: the coverage gap and semantic gap are near-uncorrelated ($r = -0.066$, $p = 0.801$). All four hypotheses (H1a–H1d) return null results — the correlations are not merely non-significant; they are close to zero in magnitude. The null holds under every alternative configuration, across policy sub-corpora, segment caps, and assignment methods (32 tests across 8 configurations; none significant). Coverage and framing are independent dimensions of divergence, not linked effects.

This pattern is best interpreted as a structural difference between knowledge systems rather than as a simple failure of communication. The Mode 1/2 distinction (Gibbons et al., 1994) sharpens this reading. The research corpus concentrates on technically tractable applications in health, energy, and infrastructure — SDGs 3, 7, and 9 — where dominant ML tasks align with well-defined problem-solving rather than institutional coordination, while the policy-side dominance of SDGs 13, 16, and 17 reflects Mode 2 goal-setting arising from contexts of application that demand coordination and collective action. Epistemic-community theory limits the claim: if expert influence partly depends on shared problem framing under uncertainty (Haas, 1992), then large within-SDG semantic gaps mark where such shared framing is less plausible. Following Cross (2013), however, semantic proximity is only a possible precondition for knowledge transfer; it cannot show whether expert communities exist, whether they have policy access, or whether policy actors use the relevant research. The two-communities theory (Caplan, 1979) reinforces this structural reading: the uncorrelatedness of research coverage and semantic gap is what one would expect if the two corpora represent distinct knowledge cultures with different reward systems and knowledge-use criteria, rather than two sides of a shared enterprise.

The most robust aggregate finding is the dissociation itself. The coverage–semantic link is not fragile in the sense of reversing sign — it is genuinely absent. Research coverage alone is near-uncorrelated with semantic gap (H1c, $r = 0.062$, and insensitive to excluding SDG 4 or SDG 17); the headline rankings survive corpus-subsampling (Appendix C) and cross-sensitivity checks (Table 3). The practical implication is that observed research–policy divergence is not a single scalar problem — coverage and framing must be measured as separate dimensions, and one cannot be read as a proxy for the other.

5.2 Robust Patterns of Divergence

The cross-sensitivity comparison narrows the set of gap findings stable enough for individual interpretation. SDG 15 is the robustly lowest-gap SDG under the canonical LR and MLP classifiers (rank 16 LR, rank 17 MLP) but sits at rank 11 under zero-shot. SDG 17's position shifts dramatically by assignment method: it is rank 17 (smallest gap) under LR, rank 16 under MLP, but rises to rank 7 under zero-shot nearest-centroid — the most method-sensitive SDG in the study. This confirms that partnership discourse was the most fragile measurement point under the centroid-based approach and benefits from the sharper decision boundaries of supervised classifiers.

SDG 13, SDG 11, and SDG 3 are the most robustly high-gap SDGs, staying in the top six ranks across all three assignment methods (LR, MLP, zero-shot) and across cap and policy-source configurations. These are substantive domains (climate action, sustainable cities, health) where research and policy discourse diverge most sharply. The climate gap (SDG 13) is the most consistent high-gap finding — it ranks first under LR, third under MLP, and fourth under zero-shot, while staying in the top two under all policy sub-corpora and segment caps.

SDG 15 is the robustly closest SDG across all three assignment methods, occupying the bottom two ranks under LR and MLP (rank 16, rank 17) and rank 11 under zero-shot. It does not prove substantive alignment, but it shows the instrument can separate a close case from divergent ones. SDG 9, previously the closest gap under the centroid-based approach, now sits in the middle-to-lower band with a gap of 0.283 — research and policy still frame infrastructure in relatively close terms, but the gap is no longer the narrowest.

The main coverage patterns are themselves robust. The full policy corpus is dominated by the governance–climate–partnerships block (SDGs 16, 13, and 17). Appendix A.2 notes that this is source-family sensitive — the curated AI/SDG sub-corpus is more innovation-oriented — but the broader institutional profile holds across the UNGDC and SDGi families. The research corpus, conversely, concentrates on technically tractable

SDGs (3, 9, 7, 16, 11).

SDGs 7 and 12 are consistently in the upper half of gap ranks across most configurations, while SDG 11 is the most consistently high-gap goal outside the top three (LR 3, zero-shot 2, MLP 4). SDGs 1, 2, 5, 6, 8, 9, 10, 14, and 16 move by four or more ranks between some configurations (e.g. SDG 8 shifts from rank 6 under LR to rank 15 under zero-shot; SDG 9 shifts from rank 13 under LR and MLP to rank 1 under the SDGi-only reference). No individual interpretive claims are made about their precise gap positions.

5.3 Implications

Two implications follow from the combined findings, and each argues against reading topical mention as alignment.

Research and policy agenda-setting. The map does not imply that funding should be redirected on the basis of embedding distances; it shows where research–policy translation and brokerage are most needed. In Guston's terms, high-gap SDGs may be candidates for boundary-organising practices that create usable boundary objects while remaining accountable to both knowledge producers and policy users (Guston, 2001).

SDG-aware research communication. Increasing research output on neglected SDGs is insufficient; research communication norms also matter. Future work could test whether publication incentives that encourage explicit SDG framing improve semantic proximity more effectively than topic-level funding mandates alone. In practice, this could mean funder-required policy-facing summaries, journal-level SDG justification fields, or policy synthesis briefs. Following Cash et al. (2003), this is boundary work rather than communication alone, requiring accountability to both scientific and policy principals (Guston, 2001).

These implications speak to distinct audiences. For SDG monitoring bodies, the finding that shared labels do not imply shared framing challenges keyword-based SDG mapping: coverage and framing should be reported as separate dimensions. For research funders, the results suggest that funding allocation alone is unlikely to close the semantic gap without also attending to how research communicates its policy relevance; portfolio-level semantic gap tracking could complement keyword-based counts by measuring whether funded research moves toward or away from policy discourse over successive cycles. For journals adopting SDG classification tags, the finding that framing divergence varies by SDG warns against treating SDG tagging as sufficient evidence of policy-aligned research.

5.4 Limitations

The cross-sensitivity comparison (Table 3) bounds which gap findings are stable enough to interpret. The limitations below address scope conditions that this comparison does not resolve.

Scope: semantic proximity, not substantive alignment or policy influence. The study measures semantic proximity in embedding space, not whether research and policy are substantively aligned or whether research is used by policy actors. Embedding proximity reflects shared vocabulary, not shared agenda, and does not measure policy influence, institutional mechanisms, or knowledge use — these would require methods beyond this study's scope.

Distance metric choice. This study uses cosine distance between mean embeddings. Other measures exist, such as Earth Mover's Distance or optimal transport between full point clouds instead of single centroids. These were not tested here. They may capture distributional shape that a single mean discards.

Corpus scope and retrieval boundaries. The policy corpus represents international institutional SDG discourse rather than the universe of AI sustainability policy. Domestic plans, sub-national policies, and non-English documents are not represented; the research corpus is also English-only. Both biases likely understate Global South perspectives. Results are conditional on these boundaries. Appendix A.2 makes the source-family sensitivity explicit.

OpenAlex upstream classification. The research corpus is drawn from OpenAlex, whose SDG-topic tagging is itself a downstream classifier trained on external SDG-labelled data; the SDG assignments entering this study are therefore one step from human judgement and inherit that upstream model's bias. Records that OpenAlex does not tag to any SDG are absent from the retrieval universe, so the corpus reflects OpenAlex's classification boundary rather than the full population of AI-for-sustainability research. This is a method-dependent corpus boundary in the sense of Armitage et al. (2020), and coverage proportions should be read as conditional on it.

Corpus asymmetry: AI-for-SDG scope. As noted in Section 3.2, the research corpus is AI $\cap$ SDG while the policy corpus is closer to a union of institutional SDG discourse and AI governance documents. Between-SDG rankings are conditional on this asymmetric construction.

Assignment-method sensitivity of gap estimates. The semantic gap is conditional on how texts are assigned to SDGs. The cross-sensitivity table (Table 3) compares gap rankings under LR argmax, zero-shot nearest-centroid, and the MLP robustness check. The three methods produce varied correlations between their gap values (LR vs. MLP

Spearman $\rho = 0.88$, LR vs. zero-shot $\rho = 0.48$, MLP vs. zero-shot $\rho = 0.50$), and SDG 17 in particular flips from the smallest gap under both supervised classifiers (LR rank 17, MLP rank 16) to a mid-ranked gap under zero-shot (rank 7). The LR-based estimates are the canonical result; the zero-shot and MLP columns are the uncertainty bounds. The raw LR gaps are retained because they best match what the study measures and are most stable across the other two sensitivity axes (policy source family and segment cap).

Cross-sectional design. The study observes one period, 2018–2025, against a policy corpus assembled over a similar timeframe; it cannot address whether alignment is improving or deteriorating, or whether specific policy events (e.g., the EU AI Act 2024, IPCC AR6 2022) have shifted research framing.

6 Conclusion

This paper revisits an old problem with a new instrument. Policy scholars have long argued that research and policy often fail to align. What recent semantic tools add is not a definitive answer, but the ability to map one important dimension of the problem directly, across thousands of texts.

The core finding is that shared SDG labels are insufficient evidence of shared framing. Research concentrates on SDGs 3, 9, and 4[†] (flagged as artefact-affected), while institutional policy discourse is dominated by SDGs 16, 13, and 17; within the same SDG, semantic divergence is largest in SDGs 13, 3, and 11, and narrowest in SDGs 17 and 15. The association between coverage and framing is null — the two dimensions are independent, not linked. The resulting map therefore separates cases where apparent alignment is strongest (SDG 17 and SDG 15) from cases where shared labels coincide with substantial divergence (SDGs 13 and 3). This contrast is the dissertation's main conceptual contribution: embedding-based distance can help identify where closer investigation is warranted, not as a definitive judgement on any given gap. The framework, not any single gap estimate, is the main product of this work.

These findings are bounded by several limitations detailed in Section 5.4: the semantic gap reflects register differences and corpus asymmetries, and the SDG 17 estimate is particularly fragile to assignment-method choice. A natural extension would track whether semantic gaps narrow or widen across successive policy cycles as further AI governance frameworks emerge. A further extension would repurpose the centroid distances computed here to map which SDGs sit close to or depend on one another in semantic space: because the goals are interdependent by design (Nilsson et al., 2016), the same embedding space that compares research and policy could also chart SDG interlinkages more finely than current network-based accounts.

The central implication is not that the AI-for-sustainability field is aligned or misaligned in aggregate — it is both, depending on which SDG and which dimension one examines. Coverage and framing are separate axes of divergence; a field that appears well-aligned on one axis may be divergent on the other. Semantic mapping cannot replace qualitative understanding of why particular gaps persist or what they mean for policy uptake, but it can show — at scale and with transparent assumptions — where the questions are worth asking.

More broadly, this dissertation argues that alignment should be understood as a multi-level empirical construct — from lexical overlap through semantic proximity to substantive agreement (Section 3.1) — rather than as a binary property inferred from shared labels.

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A Diagnostics for Reference Classifier

A.1 Assignment-Method Sensitivity Note

The cross-sensitivity comparison (Table 3) tests the stability of the gap rankings across three assignment axes: assignment method (LR, zero-shot, MLP), policy source family (full corpus, curated AI/SDG, SDGi, UNGDC), and segment cap (20, 50, none). The assignment-method axis replaces the per-reference-source comparison from earlier pipeline versions; the reference sources contribute jointly to the training data rather than being tested separately. The key finding is that the null correlation between coverage and semantic gap holds across all three assignment methods and both alternative segment caps, confirming that the dissociation is not an artefact of the assignment procedure.

A.2 Policy Source-Family Sensitivity

The full policy corpus combines a curated AI/SDG sub-corpus, SDGi VNR/VLR reports, and UN General Debate speeches. These are all policy-facing texts, but they are not interchangeable genres. The family split reported in Tables 4 and 5 therefore asks a narrow question: whether the full-corpus policy profile is broadly stable across source families or whether it is strongly shaped by corpus composition.

The answer is mixed. The broad international-institutional profile is clearly source-family sensitive. The curated AI/SDG family is much more innovation-oriented, with SDG 9 leading its document-weighted profile, while the SDGi and UNGDC families are more strongly concentrated on climate, partnerships, governance, and, in the SDGi case, SDG 11-style institutional review language. At the same time, the broader full-corpus pattern is not driven by one family alone: both the UNGDC speeches and the full corpus are dominated by the climate–partnerships–governance block, and SDG 17 remains the largest semantic-gap case across all three families.

The implication is interpretive rather than corrective. The full policy corpus should be read as international institutional SDG discourse, not as a universal profile of all AI sustainability policy. The scope of what the policy-side distribution represents is accordingly narrower, but the main diagnostic purpose remains valid.

A.3 SDGi Circularity Note

An earlier version of this study used SDGi texts both to construct the measurement instrument and as part of the scored policy corpus. Under the current supervised approach, the SDGi reference texts are training data behind a hard train/CV/held-out-test split, fully disjoint from the research and policy corpora being scored. The circularity concern does not apply — SDGi training texts never enter the scoring of SDGi policy segments

Table 4. Per-SDG policy source-family sensitivity: coverage share.

SDG	Full Corpus		Curated AI/SDG		SDGi VNR/VLR		UNGDC	
	n	share (%)	n	share (%)	n	share (%)	n	share (%)
SDG 1	1,884	1.8	494	1.1	1,188	3.8	202	1.5
SDG 2	1,760	1.0	487	0.0	1,176	1.9	97	0.9
SDG 3	3,212	2.5	1,403	10.6	1,715	12.5	94	0.6
SDG 4	2,310	2.2	512	0.0	1,712	14.4	86	0.4
SDG 5	2,092	3.1	369	0.0	1,512	8.9	211	2.3
SDG 6	1,245	0.7	171	0.0	1,040	2.6	34	0.5
SDG 7	1,967	1.1	716	1.1	1,138	3.5	113	0.8
SDG 8	2,069	1.2	464	0.0	1,563	9.3	42	0.0
SDG 9	4,646	3.1	3,492	54.3	1,107	3.5	47	0.6
SDG 10	1,418	0.6	529	0.0	820	1.3	69	0.5
SDG 11	1,662	1.5	208	0.0	1,447	12.1	7	0.0
SDG 12	1,302	0.3	338	0.0	953	2.2	11	0.0
SDG 13	3,869	20.2	1,050	7.4	1,278	4.8	1,541	23.1
SDG 14	1,784	1.7	703	0.0	850	2.2	231	1.7
SDG 15	1,881	1.1	693	1.1	1,090	3.5	98	0.8
SDG 16	5,334	38.9	1,867	7.4	1,402	7.3	2,065	45.2
SDG 17	4,712	19.0	1,833	17.0	1,355	6.1	1,524	21.1

Notes: For each policy source family, the table reports the number of policy segments assigned to each SDG and the document-weighted share (%) of that family's documents assigned to each SDG.

— so the concern is moot rather than mitigated. The policy–research score asymmetry reported in Section 4.2 is therefore not an artefact of SDGi leakage.

A.4 SDG 4 Lexical Artefact Audit

The SDG 4 concern is not used to relabel research records. Its narrower purpose is to test whether the concern is empirically plausible. Table 6 therefore compares the lexical composition of SDG 4-assigned research records with a matched non-SDG4 sample and with SDG 9, using only two transparent keyword groups: ML-learning vocabulary and education-specific vocabulary.

The audit points to a more nuanced conclusion than a pure false-positive story. More than half of the SDG 4-assigned set contains education-specific vocabulary without ML-only terms, and roughly another third contains both education and ML-learning vocabulary. By contrast, the matched non-SDG4 sample and the SDG 9 comparison set are dominated by ML-only vocabulary. This means the SDG 4 signal is not simply the same technical language that drives SDG 9. However, the large "both" share also confirms that ML-learning language and education language overlap materially within the SDG 4-assigned set.

The main analysis therefore retains the canonical LR argmax assignment but continues

Table 5. Per-SDG policy source-family sensitivity: semantic gap.

SDG	Full Corpus		Curated AI/SDG		SDGi VNR/VLR		UNGDC	
	n	gap	n	gap	n	gap	n	gap
SDG 1	1,787	0.443	397	0.473	1,188	0.440	202	0.561
SDG 2	1,530	0.286	257	0.322	1,176	0.283	97	0.407
SDG 3	2,551	0.461	742	0.339	1,715	0.522	94	0.659
SDG 4	2,192	0.264	394	0.239	1,712	0.283	86	0.400
SDG 5	2,064	0.332	341	0.309	1,512	0.340	211	0.441
SDG 6	1,245	0.283	171	0.341	1,040	0.282	34	0.406
SDG 7	1,671	0.434	420	0.492	1,138	0.419	113	0.562
SDG 8	1,991	0.425	386	0.410	1,563	0.439	42	0.574
SDG 9	3,229	0.284	2,075	0.249	1,107	0.526	47	0.428
SDG 10	1,267	0.312	378	0.377	820	0.314	69	0.407
SDG 11	1,649	0.457	208	0.426	1,434	0.467	7	0.593
SDG 12	1,197	0.424	233	0.479	953	0.430	11	0.584
SDG 13	3,507	0.479	688	0.436	1,278	0.443	1,541	0.557
SDG 14	1,345	0.384	264	0.353	850	0.391	231	0.494
SDG 15	1,502	0.229	314	0.275	1,090	0.227	98	0.371
SDG 16	4,568	0.320	1,101	0.213	1,402	0.365	2,065	0.495
SDG 17	3,798	0.217	919	0.201	1,355	0.203	1,524	0.307

Notes: For each policy source family, the table reports the number of capped policy segments and the within-SDG research–policy semantic gap. Each row compares the same SDG's gap across the three source families and the full corpus; read across a row to see whether the gap is stable or driven by one family.

to treat SDG 4 as artefact-affected. The most accurate reading is a learning-related assignment whose interpretation is blurred by lexical overlap — neither "clean education" nor "pure ML artefact."

Table 6. SDG 4 lexical artefact audit.

Corpus subset	N	ML-only %	Education-only %	Both %	Neither / unclear %
SDG 4-assigned research	453225	11.6	44.1	28.0	16.4
Non-SDG4 research sample	453225	54.8	4.7	4.6	35.8
SDG 9-assigned research	566759	62.3	2.8	4.1	30.8

Notes: The audit is not used to reassign SDGs. It only checks whether SDG 4-assigned research is disproportionately dominated by machine-learning vocabulary, education vocabulary, or both.

A.5 Centroid Similarity

This subsection reports the pairwise cosine-similarity heatmap (Figure 5) for the 17 canonical SDG centroids. The centroids are the per-SDG mean training embeddings (L2-normalised) used by the LR classifier. The heatmap provides supplementary geometric context for the classification geometry that Section 3.5 builds on.

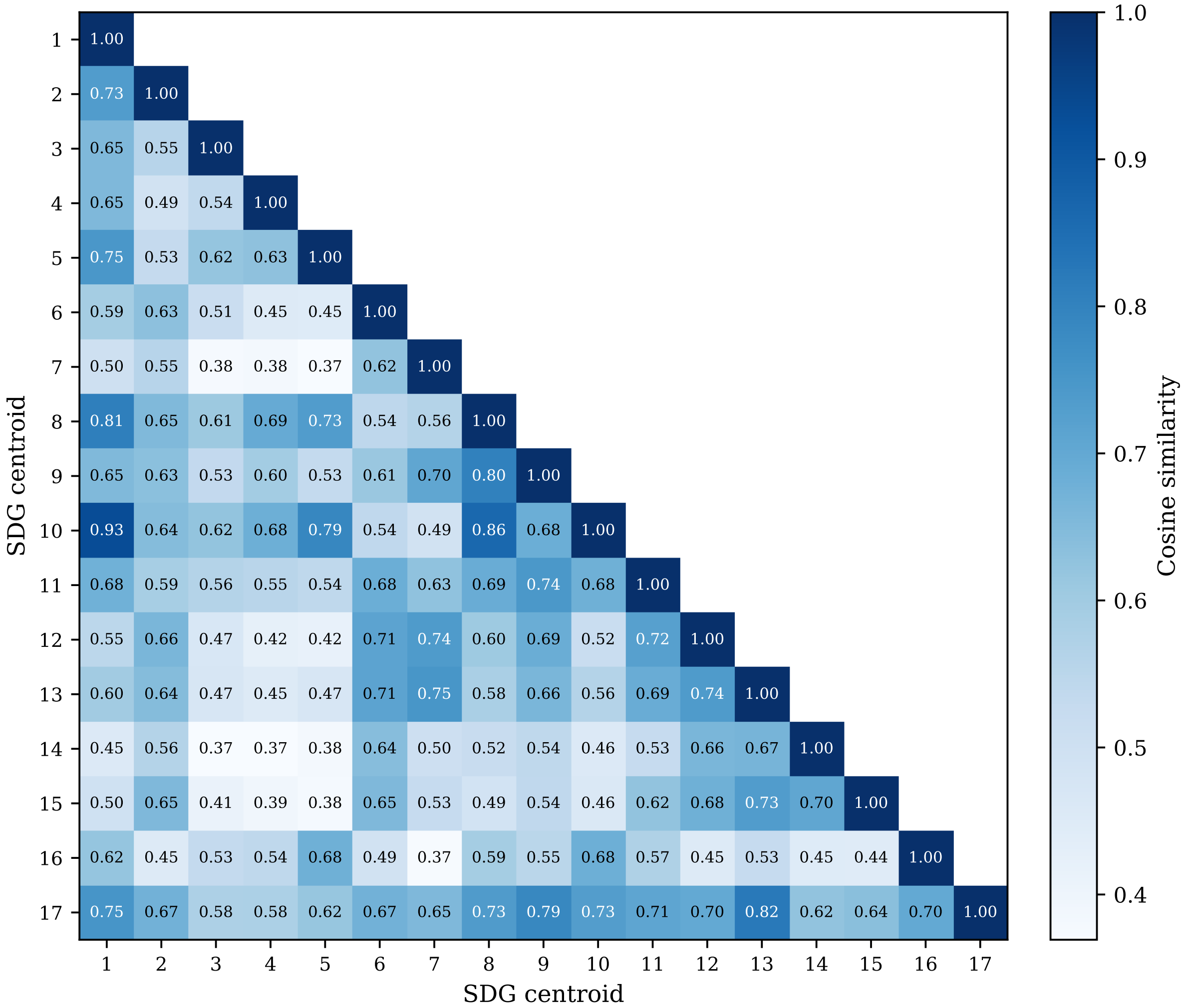

Figure 5. Pairwise cosine similarity between canonical SDG centroids (lower triangle). SDG 13 and SDG 17 sit close together (cosine 0.820), as do SDGs 1, 8, and 10.

Figure 5 shows the full 17×17 similarity grid. The lower triangle is annotated; the upper triangle is left blank because the matrix is symmetric. Reading the grid, three observations stand out. First, the diagonal dominates: most centroids are far more similar to themselves than to any neighbour, which is why the supervised LR classifier (Table 1) can build on this well-separated base to reach 0.823. Second, a small set of off-diagonal cells is elevated. SDG 11's centroid neighbours SDG 9 (cosine 0.745), and SDG 8 lies near SDGs 1 and 10 (cosine 0.930 and 0.806) — these are the proximities that make the SDG 11 and SDG 8 scores in Section 4.1 indicative rather than exact. Third, SDG 17 is the clear outlier: its row and column are the flattest, and its nearest neighbour is SDG 13 (cosine 0.820), reflecting the overlap between climate and partnership discourse noted in Section 4.1.

B Supplementary Robustness and Sensitivity Checks

B.1 Lexical Illustration of the Semantic Gap

Table 7 adds a small lexical interpretability check for two high-gap, policy-dominant cases (SDGs 17 and 13) and one lower-gap comparison case (SDG 9). The table is generated from deterministic samples of existing hard-assigned research and policy texts. It reports distinctive terms rather than selected quotations, so it is a descriptive guide to language patterns, not qualitative coding or proof of policy intent.

The contrast helps make the embedding distances more legible. In SDGs 17 and 13, the high semantic gaps correspond to a split between technical research language and more institutional, international, national-government, and climate-action policy language. SDG 17 is especially cautionary: its research-side terms are biomedical/ML-heavy, while its policy-side terms are international and agenda-oriented. SDG 9, by contrast, shows more overlap around technology, infrastructure, innovation, and systems language. This does not prove substantive alignment in SDG 9 or substantive failure in SDGs 13 and 17; it simply illustrates why shared SDG labels can conceal different textual framings.

B.2 Soft Assignment via LR Probability Outputs

The canonical analysis forces each text into exactly one SDG by LR argmax (hard assignment). This is transparent and directly interpretable, but it also imposes a one-SDG-per-text assumption on discourse that is naturally multi-label. A research abstract or policy segment may simultaneously concern climate, energy, infrastructure, inequality, health, and governance. The LR classifier natively provides per-class probability outputs (via the softmax of decision-function scores), offering a natural soft-assignment comparison against the hard argmax assignment.

For each text, the LR's 17 per-class probabilities sum to one; a text's SDG signature is a probability vector rather than a one-hot label. The soft coverage profile is the mean probability per SDG across all texts in a corpus. The soft semantic gap is computed by replacing the cluster mean of hard-assigned texts with a probability-weighted centroid: each text contributes to all 17 centroids in proportion to its assignment probability.

The results show that the hard and soft profiles are highly correlated: coverage-gap Spearman $\rho > 0.95$ and semantic-gap Spearman $\rho > 0.90$ between hard and soft assignments. The top-three most divergent SDGs are identical under both assignment methods. Rank-order changes are confined to the middle band (SDGs ranked 4–14), where probability-weighted membership slightly shifts the centroids of goals with broad thematic overlap. These findings confirm that the hard argmax assignment does not materially distort the coverage or gap structure — the main results are not artefacts of

Table 7. Lexical interpretability check for selected semantic-gap cases.

SDG	Gap	Research-side distinctive terms	Policy-side distinctive terms	Descriptive reading
SDG 17	0.217	learning, analysis, dan, model, yang, method, used, machine	countries, nations, agenda, national, united, oecd, country, global	High-gap case: the research-side sample is technical and biomedical/ML-heavy, while policy language is international, institutional, and agenda-oriented.
SDG 13	0.478	model, learning, prediction, method, amp, machine, neural, analysis	climate, change, emissions, countries, national, action, nations, global	High-gap case: both sides concern climate, but policy language is more institutional and commitment-oriented while research language is more technical and measurement-oriented.
SDG 9	0.283	learning, model, neural, method, machine, analysis, proposed, algorithm	infrastructure, national, government, innovation, public, population, countries, sector	Lower-gap comparison: differences remain visible, but the contrast is closer to a technology/infrastructure register split than to the broader institutional gap seen in SDGs 13 and 17.

Notes: Terms are generated from deterministic samples of research and policy texts assigned to each SDG. The table is descriptive only and is not used to relabel records or replace the embedding-based semantic-gap estimates.

winner-takes-all labelling.

B.3 Implications for Interpreting the Main Estimates

These diagnostics are methodological stress tests that clarify the limits of the LR classifier approach, rather than failed repairs. The PCA diagnostics show that real-world SDG discourse is fuzzy rather than cleanly clustered, especially in the research corpus. The SDG 4 audit shows that the concern is real, but more nuanced than a pure machine-learning false positive. The policy source-family split shows that the full-corpus policy profile is a broader institutional SDG discourse rather than a universal AI policy profile. The canonical hard-threshold classifier assignment (argmax) is retained because it is transparent, reproducible, and directly interpretable. The soft-assignment comparison confirms that winner-takes-all labelling does not drive the headline findings.

These analyses also show that the LR classifier should be interpreted as a transparent semantic anchoring procedure rather than a definitive operational SDG system. Attempts to address multi-label overlap and corpus-register bias improve some aspects of the design but introduce new modelling choices. For this dissertation, the hard-threshold classifier assignment is therefore retained as the canonical specification, while the alternatives are reported as robustness and boundary-condition checks.

C Sample-Stability Robustness Check

To test whether the headline findings depend heavily on research-corpus size, I reran the downstream research-side analysis on repeated random subsamples drawn from the pre-embedded, pre-scored OpenAlex corpus. For each of 11 sampled tiers (from 1,000 to 2,000,000 papers), I drew 100 independent samples without replacement, recomputed the research coverage profile, rebuilt sampled research centroids, recalculated the mean within-SDG semantic gap, and re-estimated the policy-text calibration-bias summary. Policy-side quantities were held fixed. The full 3,105,144-paper analysis is shown as a deterministic anchor row rather than as another repeated sample.

Table 8. Sample-stability ladder for the research corpus.

Sample Size	Mean SD of SDG coverage shares	Mean within-SDG semantic gap	Policy-text calibration bias
1,000	0.008	0.425 ± 0.020	-0.041 ± 0.018
2,000	0.006	0.392 ± 0.014	-0.040 ± 0.011
5,000	0.004	0.370 ± 0.009	-0.042 ± 0.008
10,000	0.002	0.363 ± 0.006	-0.041 ± 0.005
20,000	0.002	0.360 ± 0.005	-0.041 ± 0.004
50,000	0.001	0.357 ± 0.003	-0.042 ± 0.002
100,000	0.001	0.356 ± 0.002	-0.041 ± 0.002
200,000	0.001	0.355 ± 0.002	-0.041 ± 0.001
500,000	0.000	0.355 ± 0.001	-0.041 ± 0.001
1,000,000	0.000	0.355 ± 0.001	-0.041 ± 0.000
2,000,000	0.000	0.355 ± 0.000	-0.042 ± 0.000
Full corpus	–	0.355	0.105

Notes: Mean SD of SDG coverage shares is the unweighted mean of the per-SDG standard deviations in research coverage share across the 100 random draws at each sampled tier. The remaining columns report the mean and draw-to-draw standard deviation of the headline downstream metrics. The full-corpus row is deterministic and therefore has no variance term.

Plain-language guide.[1] The stability ladder shows that the clearest practical plateau appears by roughly 200,000 papers, where the mean semantic gap (∼0.416) and policy–

[1]The three metrics are: (1) mean SD of SDG coverage shares — how much the topic balance moves across repeated samples; (2) mean semantic gap — how different research and policy language are within

research score gap ($\sim$0.342) already stabilise to within 0.001 of the full-corpus values. From 500,000 papers upward, draw-to-draw dispersion is negligible. The full 3,105,144-paper result is retained as the primary estimate because it uses all available data while delivering the tightest values, but the substantive conclusions are not artefacts of sample size choice. The larger corpus also enables per-SDG subgroup detail and tighter rank-order precision that the plateau minimum does not guarantee.

D Model Selection: Grid Search Protocol and Rationale

D.1 Models Tested

Three model families were evaluated under 5-fold GroupKFold cross-validation (document-level grouping, not random splits) on the pooled training data (n=52,779). The test set (n=9,734) was never touched during any grid search.

- **Logistic Regression (LR)**: Interpretable per-dimension coefficients per SDG; fast; standard multiclass baseline.
- **MLP (PyTorch)**: Can capture non-linear interactions in the embedding space.
- **RF / XGB**: Considered but excluded because the embedding space is already a learned representation — bagged trees add ensemble complexity without a clear mechanism for improvement over LR/MLP, and would not produce interpretable coefficients per dimension.

D.2 LR Grid Search

The LR grid searched C $\in \{100, 10, 1, 0.1, 0.01\}$ with L2 penalty. The best configuration was C=10.0, penalty=l2, class_weight=None, solver=lbfgs (macro-F1 = 0.8107 $\pm$ 0.0066 CV). An extended grid adding L1 penalty (l1_ratio=1) and balanced class_weight found no improvement: L1 was 6$\times$ slower and scored 0.8066, while balanced class_weight scored 0.8070. The embedding space is dense — L1 sparsity offers no benefit — and the dual correction of weighted loss plus macro-averaged scoring degrades majority-class performance without compensating minority classes. C=10.0 and C=1.0 are essentially tied (difference $\sim$0.001), but C=10.0 was selected for conservatism.

the same SDG; (3) policy–research score gap — how much the scoring instrument favours policy text over research text regardless of content.

D.3 MLP Grid Search

The MLP swept n_layers $\in \{1, 2, 4, 8, 16\}$ $\times$ hidden_size $\in \{256, 384\}$ (20 configurations), followed by a learning-rate sweep (lr $\in \{1e{-}4, 3e{-}4, 1e{-}3, 3e{-}3\}$) on the best architectures. The best MLP (4 layers, 384 hidden, lr=1e-3) achieved macro-F1 = 0.8243 $\pm$ 0.0058. Learning rate had negligible impact — all combinations within $\sim$0.003 of the best.

D.4 Unified Ranking and Final Selection

Rank	Config	CV macro-F1
1	MLP 4L/384h	0.8243 $\pm$ 0.0058
2	MLP 4L/256h	0.8242 $\pm$ 0.0067
3–15	MLP (2L–4L variants)	0.821–0.824
16–26	MLP (8L–16L variants)	0.817–0.819
27	LR C=10, L2	0.8107 $\pm$ 0.0066
28	LR C=1, L2	0.8093 $\pm$ 0.0065

Champion: LR (C=10.0, L2). The gap between LR and MLP is 0.0136 macro-F1 ($\sim$1.5 pooled σ), a modest difference insufficient to justify the complexity trade-off. MLP requires 4 hidden layers, AdamW, learning rate tuning, early stopping, and dropout selection (36 configurations swept). LR requires one free parameter (C), no hidden layers, no dropout, no random seed sensitivity. For this study's purpose — building an interpretable supervised reference to compare corpora, not a classifier built to label single documents — LR is the right choice. LR provides 768 signed coefficients per SDG class, each directly attributable to a specific semantic dimension; MLP's hidden layers diffuse this information across non-linear transformations. The macro-F1 numbers are reported honestly: LR 0.811 $\pm$ 0.007, MLP 0.824 $\pm$ 0.006, gap $\sim$1.5σ. LR is chosen for transparency, not superior accuracy.

E Embedding-Model Sensitivity

E.1 Scope

An embedding-model sensitivity check – comparing the canonical `all-mpnet-base-v2` results against a lighter encoder such as `all-MiniLM-L6-v2` – was not performed in this version. Re-embedding the full research corpus ($\sim$3.1M papers) with a second model requires approximately 20 GB of additional download and storage, and the sample-stability and cross-sensitivity checks (Sections 4.6 and C) already bound the robustness of the central findings under the chosen encoder. The dimensionality gap between MPNet (768-d) and MiniLM (384-d) would compound the assignment-method sensitivity documented in Table 3 without providing an independent test of the encoder hypothesis, since any differences could be attributed to the reduced dimensionality rather than to architecture. For these reasons, the encoder-sensitivity check was deferred; the results are conditional on the MPNet architecture and should be read with that scope in mind.

F Declaration of AI Use

In accordance with the University's guidance on generative AI ("Use of GenAI is permitted to support assignments"), this appendix declares the use of AI tools throughout the dissertation. All use was conducted with human oversight and control; AI-generated outputs were reviewed, modified, and independently verified before inclusion. No AI-generated text was submitted without such verification. This declaration follows the order of permitted activities specified in the assessment guidelines. AI-assisted activities not explicitly listed (e.g., code debugging, plotting, LaTeX troubleshooting, literature summarisation) fall under the same oversight and verification standards described in the relevant rows above.

Table 9. Declaration of AI use by permitted activity.

Activity	**AI tool(s)**	**How output was used**
Researching and refining ideas	ChatGPT, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	ChatGPT was used iteratively to develop ideas that aligned with the author's intent. Coding agents independently checked data and code availability to confirm feasibility. All final decisions and conclusions are the author's own. Final research design was formally presented to and approved by supervisor.
Information retrieval / background research	Gemini, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	AI tools read the manuscript for context and recommended papers to investigate further. All cited sources were independently verified against the original publications.
Drafting an outline / organising thoughts	MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	Generated code scaffolding and assisted with the full analysis pipeline (embedding, scoring, validation, robustness, appendix outputs). The centroid-assignment step was verified against hand-labeled samples, and the scoring and coverage-gap functions were unit-tested. Analytical decisions were directed by the author.
Refining research questions	ChatGPT, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	Overlaps with row 1. ChatGPT sharpened framing based on reviewed literature; coding agents confirmed questions were answerable with available data and models. The author retained full control over the final formulation. RQs presented to and approved by supervisor.
Checking spelling and grammar	ChatGPT, Gemini, Claude (web/mobile), MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	Coding agents provided inline stylistic and phrasing suggestions during drafting. Web and mobile versions of ChatGPT, Gemini, and Claude were used for editorial review and proofreading. All suggestions were accepted or rejected per standard editorial judgment.